

Azure DevOps Assignment

Banking Application Scenario

PART 1 - Work Item Breakdown

Epic 1: User Management

Feature 1.1: User Registration & Authentication

- User Story 1.1.1: New User Registration
As a new user, I want to register using my email and password so I can access banking services.
 - Task: Design registration form (UI)
 - Task: Implement registration API
 - Task: Store user credentials securely in the database
 - User Story 1.1.2: User Login
As a returning user, I want to log in securely so I can access my account.
 - Task: Create login form
 - Task: Implement login API with session/token
 - Task: Validate credentials and handle errors
-

Feature 1.2: Profile Management

- User Story 1.2.1: Edit User Profile
As a user, I want to update my personal information to keep my account up to date.
 - Task: Build editable profile form
 - Task: Connect form to update profile API
 - Task: Validate input fields before submission
-

Epic 2: Account Services

Feature 2.1: Account Creation

- User Story 2.1.1: Open New Account
As a user, I want to create a savings or checking account to manage my money.
 - Task: Design account creation form

- Task: Implement backend logic to open an account
 - Task: Store account details in the database
-

Feature 2.2: View Account Balance

- User Story 2.2.1: Check Account Balance
As a user, I want to view my current balance so I can track my funds.
 - Task: Create balance retrieval API
 - Task: Display balance on dashboard
 - Task: Handle loading and error states
-

Epic 3: Fund Transfer

Feature 3.1: Transfer Funds

- User Story 3.1.1: Transfer to Another Account
As a user, I want to transfer money to another user's account securely.
 - Task: Build transfer UI form
 - Task: Create fund transfer API
 - Task: Validate transfer details (amount, recipient, balance check)
-

Epic 4: Transaction History

Feature 4.1: View Transaction History

- User Story 4.1.1: View All Transactions
As a user, I want to see a list of all past transactions to track my spending.
 - Task: Create transaction history API
 - Task: Design history view with date and amount
 - Task: Add pagination for long lists
-

Feature 4.2: Filter Transactions

- User Story 4.2.1: Filter by Date or Amount
As a user, I want to filter my transactions by date or amount to find specific entries.
 - Task: Add date and amount filter UI
 - Task: Integrate filters with API

- Task: Display filtered results in real-time

Epic 5: Security & Compliance

Feature 5.1: Multi-Factor Authentication

- User Story 5.1.1: Enable 2FA
As a user, I want to enable two-factor authentication to secure my login.
 - Task: Provide 2FA setup option in profile
 - Task: Send OTP via SMS or Email
 - Task: Validate OTP before granting access

Feature 5.2: Audit Logging

- User Story 5.2.1: Track Account Activity
As an admin, I want to track user activities to ensure system security.
 - Task: Implement backend logging of user actions
 - Task: Store logs in a secure table
 - Task: Create admin-only access to audit logs

Azure DevOps Portal View

EPIC:

The screenshot displays the Azure DevOps portal interface for a project named 'Banking Application'. The left sidebar shows navigation options: Overview, Boards, Work items (selected), Backlogs, Sprints, Queries, Delivery Plans, Analytics views, Repos, and Project settings. The main area shows a list of work items under the 'Security & Compliance' epic. A notification banner at the top indicates a transition to new IP addresses for Azure DevOps Services.

ID	Title	Assigned To	State	Area Path
144	Security & Compliance	Varshini M	New	Banking Ap
133	Transaction History	Varshini M	New	Banking Ap
127	Fund Transfer	Varshini M	New	Banking Ap
116	Account Services	Varshini M	New	Banking Ap
101	User Management	Varshini M	New	Banking Ap

Features:

As previously announced in our blog, we are transitioning to new IP addresses. If you have not done so already, please update your firewall allowlists with new IP address ranges of Azure DevOps Services. For detailed information and the latest updates, visit our [status page](#).

Work items

Recently updated | + New Work Item | Open filtered view in Queries | Column Options | ...

Filter by keyword

ID	Title	Assigned To	State	Area Path
150	Audit Logging	Varshini M	New	Banking.
145	Multi-Factor Authentication	Varshini M	New	Banking.
139	Filter Transactions	Varshini M	New	Banking.
134	View Transaction History	Varshini M	New	Banking.
128	Transfer Funds	Varshini M	New	Banking.
122	View Account Balance	Varshini M	New	Banking.
117	Account Creation	Varshini M	New	Banking.

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Work items

Recently updated | + New Work Item | Open filtered view in Queries | Column Options | ...

Filter by keyword

ID	Title	Assigned To	State	Area Path
139	Filter Transactions	Varshini M	New	Banking.
134	View Transaction History	Varshini M	New	Banking.
128	Transfer Funds	Varshini M	New	Banking.
122	View Account Balance	Varshini M	New	Banking.
117	Account Creation	Varshini M	New	Banking.
111	Profile Management	Varshini M	New	Banking.
102	User Registration & Authentication	Varshini M	New	Banking.

User Story:

BA

Banking Application

+

Overview

Boards

Work items

Boards

Backlogs

Sprints

Queries

Delivery Plans

Analytics views

Repos

Project settings

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Work items

Recently updated

+ New Work Item

Open filtered view in Queries

Column Options

...

Filter by keyword

User Story

Assigned to

States

Area

Tags

X

ID	Title	Assigned To	State	Area Path
151	Track Account Activity	Varshini M	New	Banking.
146	Enable 2FA	Varshini M	New	Banking.
140	Filter by Date or Amount	Varshini M	New	Banking.
135	View All Transactions	Varshini M	New	Banking.
129	Transfer to Another Account	Varshini M	New	Banking.
123	Check Account Balance	Varshini M	New	Banking.
118	Open New Account	Varshini M	New	Banking.

BA

Banking Application

+

Overview

Boards

Work items

Boards

Backlogs

Sprints

Queries

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Work items

Recently updated

+ New Work Item

Open filtered view in Queries

Column Options

...

Filter by keyword

User Story

Assigned to

States

Area

Tags

X

135	View All Transactions	Varshini M	New	Banking.
129	Transfer to Another Account	Varshini M	New	Banking.
123	Check Account Balance	Varshini M	New	Banking.
118	Open New Account	Varshini M	New	Banking.
112	Edit User Profile	Varshini M	New	Banking.
104	User Login	Varshini M	New	Banking.
103	New User Registration	Varshini M	New	Banking.

Tasks:

Azure DevOps varshini2108varshu / Banking Application / Boards / Work items

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Work items

Recently updated | + New Work Item | Open filtered view in Queries | Column Options | ...

Filter by keyword Task Assigned to States Area Tags X

ID	Title	Assigned To	State	Area Path
154	Create admin-only access to audit logs	Varshini M	New	Banking.
153	Store logs in a secure table	Varshini M	New	Banking.
152	Implement backend logging of user actions	Varshini M	New	Banking.
149	Validate OTP before granting access	Varshini M	New	Banking.
148	Send OTP via SMS or Email	Varshini M	New	Banking.
147	Provide 2FA setup option in profile	Varshini M	New	Banking.
143	Display filtered results in real-time	Varshini M	New	Banking.

Azure DevOps varshini2108varshu / Banking Application / Boards / Work items

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Work items

Recently updated | + New Work Item | Open filtered view in Queries | Column Options | ...

Filter by keyword Task Assigned to States Area Tags X

ID	Title	Assigned To	State	Area Path
142	Integrate filters with API	Varshini M	New	Banking.
141	Add date and amount filter UI	Varshini M	New	Banking.
138	Add pagination for long lists	Varshini M	New	Banking.
137	Design history view with date and amount	Varshini M	New	Banking.
136	Create transaction history API	Varshini M	New	Banking.
132	Validate transfer details (amount, recipient, balance check)	Varshini M	New	Banking.
131	Create fund transfer API	Unassigned	New	Banking.

Azure DevOps varshini2108varshu / Banking Application / Boards / Work items

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Work items

Recently updated | + New Work Item | Open filtered view in Queries | Column Options | ...

Filter by keyword Task Assigned to States Area Tags X

ID	Title	Assigned To	State	Area Path
130	Build transfer UI form	Varshini M	New	Banking.
126	Handle loading and error states	Varshini M	New	Banking.
125	Display balance on dashboard	Varshini M	New	Banking.
124	Create balance retrieval API	Varshini M	New	Banking.
121	Store account details in the database	Varshini M	New	Banking.
120	Implement backend logic to open an account	Varshini M	New	Banking.
119	Design account creation form	Varshini M	New	Banking.

Azure DevOps

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Banking Application

Boards

Work items

Search

VM

BA Banking Application

+

Overview

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Work items

Recently updated

+ New Work Item

Open filtered view in Queries

Column Options

...

Filter by keyword

Task

Assigned to

States

Area

Tags

X

ID	Title	Assigned To	State	Area Path
115	Validate input fields before submission	Varshini M	New	Banking.
114	Connect form to update profile API	Varshini M	New	Banking.
113	Build editable profile form	Varshini M	New	Banking.
110	Validate credentials and handle errors	Varshini M	New	Banking.
109	Implement login API with session/token	Varshini M	New	Banking.
108	Create login form	Varshini M	New	Banking.
107	Store user credentials securely in the database	Varshini M	New	Banking.

Azure DevOps

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Banking Application

Boards

Work items

Search

VM

BA Banking Application

+

Overview

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Work items

Recently updated

+ New Work Item

Open filtered view in Queries

Column Options

...

Filter by keyword

Task

Assigned to

States

Area

Tags

X

ID	Title	Assigned To	State	Area Path
113	Build editable profile form	Varshini M	New	Banking.
110	Validate credentials and handle errors	Varshini M	New	Banking.
109	Implement login API with session/token	Varshini M	New	Banking.
108	Create login form	Varshini M	New	Banking.
107	Store user credentials securely in the database	Varshini M	New	Banking.
106	Implement registration API	Varshini M	New	Banking.
105	Design registration form (UI)	Varshini M	New	Banking.

Boards:

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Banking Application Team

Board Analytics

Stories

View as backlog

New

Active 3/5

Resolved 2/5

Closed

103 New User Registration

New

Varshini M

0/3

140 Filter by Date or Amount

New

Varshini M

0/3

104 User Login

Active

Varshini M

0/3

123 Check Account Balance

Active

Varshini M

0/3

146 Enable 2FA

Active

Varshini M

0/3

112 Edit User Profile

Resolved

Varshini M

0/3

129 Transfer to Another Account

Resolved

Varshini M

0/3

151 Track Account Activity

Closed

Varshini M

0/2

135 View All Transactions

Closed

Varshini M

0/3

118 Open New Account

Closed

Varshini M

0/3

Banking Application Team

Board Analytics

Stories

View as backlog

New

Active 3/5

Resolved 2/5

Closed

+ New item

103 New User Registration

New

Varshini M

0/3

+ Add Task

Design registration form (UI)

Implement registration API

Store user credentials securely in...

140 Filter by Date or Amount

New

104 User Login

Active

Varshini M

0/3

+ Add Task

Create login form

Implement login API with session...

Validate credentials and handle er...

123 Check Account Balance

Active

Varshini M

0/3

112 Edit User Profile

Resolved

Varshini M

0/3

+ Add Task

Build editable profile form

Connect form to update profile API

Validate input fields before submi...

129 Transfer to Another Account

Resolved

Varshini M

0/3

151 Track Account Activity

Closed

Varshini M

0/2

+ Add Task

Store logs in a secure table

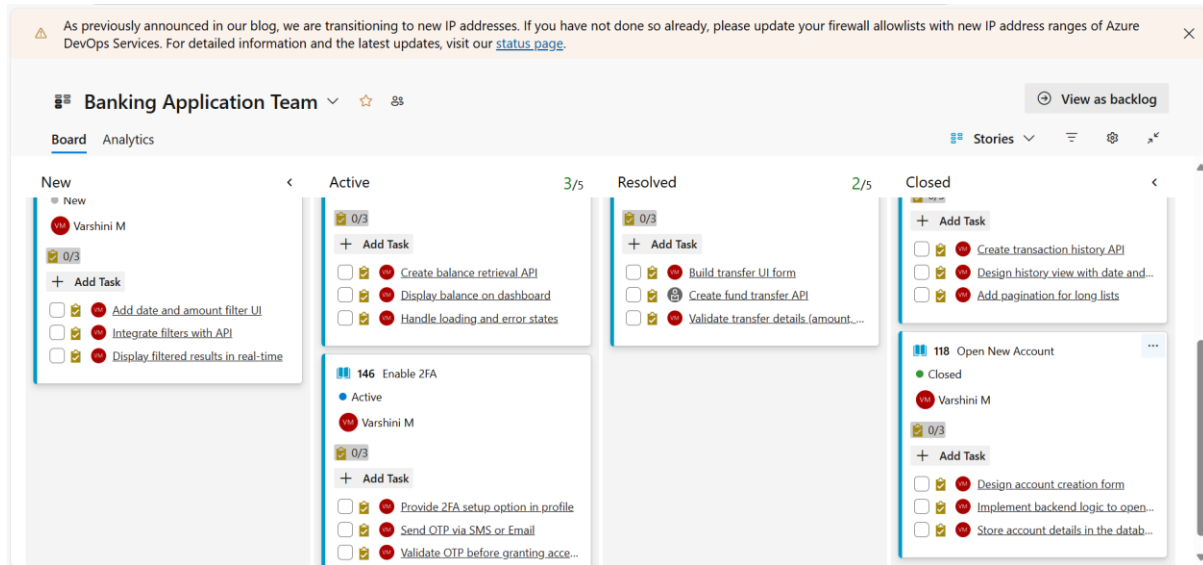
Create admin-only access to audit...

135 View All Transactions

Closed

Varshini M

0/3



PART 2 - Azure DevOps Git Setup

project structure:

Banking Application/

├── README.md

├── src/

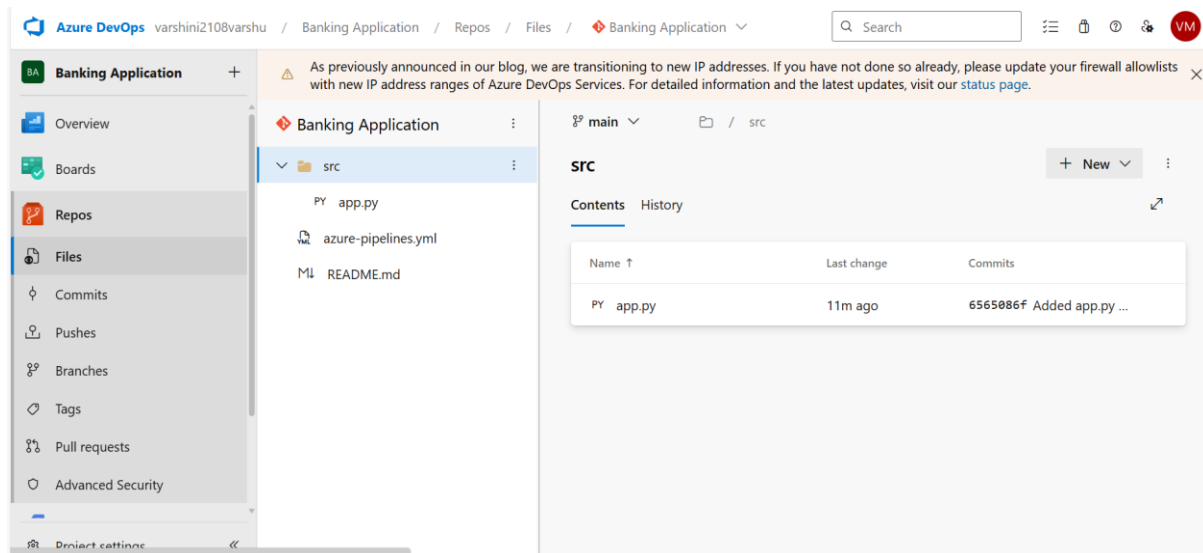
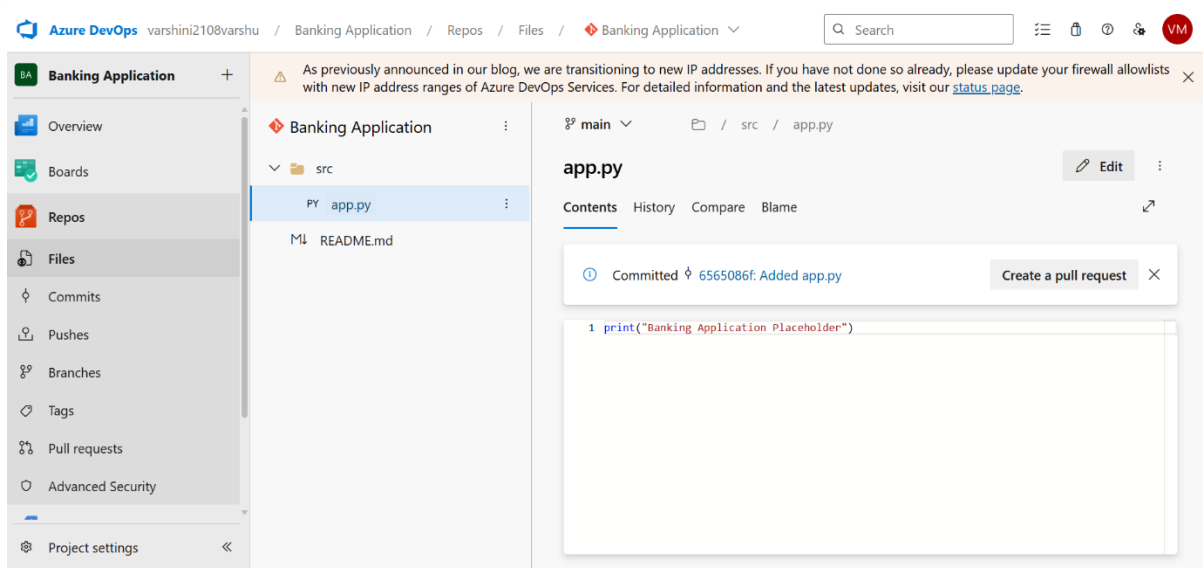
| └── app.py

└── .azure-pipelines.yml

Create a Python file(e.g. app.py)

```
#app.py
```

```
print("Banking Application Placeholder")
```



PART 3 - Configure CI/CD Pipeline

1. Create Simple Pipeline

Example: azure-pipelines.yml

trigger:

- main

pool:

vmImage: 'ubuntu-latest'

steps:

- task: UsePythonVersion@0

inputs:

versionSpec: '3.x'

- script: python src/app.py

displayName: 'Run Banking Application Placeholder'

The screenshot shows the 'Review your pipeline YAML' step in the Azure DevOps pipeline configuration. The left sidebar contains navigation links for Banking Application, Overview, Boards, Repos, Pipelines, Environments, Library, Test Plans, Artifacts, and Project settings. The main area displays the pipeline YAML code for 'Banking Application / azure-pipelines.yml'. The code defines a trigger for 'main', a pool with 'ubuntu-latest' image, and a single step named 'Run Banking Application Placeholder' using the 'UsePythonVersion@0' task with inputs '3.x' and script 'python src/app.py'. The 'Review' tab is active, and buttons for 'Variables', 'Save and run', and 'Show assistant' are visible.

```
1 trigger:
2   - main
3
4 pool:
5   - vmImage: 'ubuntu-latest'
6
7 steps:
8   Settings
9   - task: UsePythonVersion@0
10     inputs:
11       versionSpec: '3.x'
12
13   - script: python src/app.py
14     displayName: 'Run Banking Application Placeholder'
```

The screenshot shows the summary of a pipeline run. The left sidebar is identical to the previous screenshot. The main area displays the pipeline run details for '#20250707.1 • Set up CI with Azure Pipelines'. The 'Summary' tab is active, showing the repository 'Banking Application' at commit '6ea07102', the time 'Just now', and the status 'Queued'. A 'Jobs' table lists the job 'Job' with a status of 'Queued'. A 'View 5 changes' button is also present.

Name	Status	Duration
Job	Queued	

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Banking Application

Pipelines

Banking Application

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Q

Search

VM

BA

Banking Application

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Overview

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Pipelines

Pipelines

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Artifacts

Project settings

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×

✖

#20250707.1 • Set up CI with Azure Pipelines

Banking Application

Rerun failed jobs

Run new

:

ℹ

This run will be cleaned up after 1 month based on your project settings.

Summary

Code Coverage

Individual CI by

VM

 Varshini M

Repository and version

Time started and elapsed

Related

Tests and coverage

Banking Application

Just now

0 work items

[Get started](#)

main

6ea07102

<1s

0 artifacts

View 5 changes

Errors 1

✖

No hosted parallelism has been purchased or granted. To request a free parallelism grant, please fill out the following form <https://a...>

20250707.1